

**AMENDMENT NO. 1 NOVEMBER 2020**

**TO**

**IS 13656 : 2019 INTERNAL COMBUSTION ENGINE  
CRANKCASE OILS FOR AUTOMOTIVE APPLICATION  
( DIESEL AND GASOLINE ) — SPECIFICATION**

*( Third Revision )*

*(Page 3, Clause 3.4)* — Insert the following Note at the end:

‘NOTE — Multiple Test Acceptance Criteria (MTAC) — MTAC is any data-based approach for evaluation of the quality and performance of a formulation where more than one test may be run. Generally for a candidate tested once, test data for each criterion shall be a pass. For a candidate tested twice, the mean (average) value of each result shall be a pass. For a candidate tested three or more times, one test might be declared an outlier and thus discarded and the mean (average) value of retained test data for each result shall be a pass. MTAC is applicable to Table 2, Table 3, Table 4, Table 5.1, Table 6.1, Table 7.1 and Table 8.1.’

*(Page 11, Table 6.1 Engine Test Sequence for Diesel Engine Oils for EDL8, col (3), row 1)* — Substitute ‘EDL8’ for ‘EDL 7’.

*(Page 11, Table 6.1 Engine Test Sequence for Diesel Engine Oils for EDL8, col (1), row 4, Caterpillar 1R)* — Substitute the following for existing:

| Test Technique | Characteristics                                                    | Requirements<br>EDL8 |         |         | Method of Test, Ref to ASTM |
|----------------|--------------------------------------------------------------------|----------------------|---------|---------|-----------------------------|
| (1)            | (2)                                                                | (3)                  |         |         | (4)                         |
|                |                                                                    | 1 test               | 2 test  | 3 test  |                             |
| Caterpillar 1R | Initial oil consumption (IOC), (0h-252h), g/h, average, <i>Max</i> | 13.1                 | 13.1    | 13.1    | D 6923                      |
|                | Final oil consumption (FOC), (432h-504h), g/h, average, <i>Max</i> | IOC+1.8              | IOC+1.8 | IOC+1.8 |                             |

**Price Group 2**

(Page 17, Table 9.1 Engine Test Sequence for Petrol Engine Oils, row 1 to 3) — Substitute the following:

| Test Technique | Characteristics | Requirements |       |       | Method of Test, Ref to ASTM |
|----------------|-----------------|--------------|-------|-------|-----------------------------|
|                |                 | EPL 2        | EPL 3 | EPL 4 |                             |
| (1)            | (2)             | (3)          | (4)   | (5)   | (6)                         |

for

| Test Technique | Characteristics | Requirements EDL8 |        |        | Method of Test, Ref to ASTM |
|----------------|-----------------|-------------------|--------|--------|-----------------------------|
|                |                 | EPL 2             | EPL 3  | EPL 4  |                             |
| (1)            | (2)             | (3)               | (4)    | (5)    | (6)                         |
|                |                 | 1 test            | 2 test | 3 test |                             |

[Page 19, Table 10.1 Engine Test Sequence for Petrol Engine Oils for EPL 5, col (1), Last row, Sequence VIII), b)] — Substitute the following:

| Test Technique |    | Characteristics             | Primary Performance Criteria EPL 5 | Test Method |
|----------------|----|-----------------------------|------------------------------------|-------------|
| (1)            |    | (2)                         | (3)                                | (4)         |
| Sequence VIII  | b) | Shear Stability             | Refer Note 3                       | ASTM D6709  |
|                |    | or                          |                                    |             |
|                |    | Shear Stability (30 Passes) | Refer Table 11.3A                  | ASTM D6278  |

for

| Test Technique |    | Characteristics | Primary Performance Criteria EPL 5 | Test Method |
|----------------|----|-----------------|------------------------------------|-------------|
| (1)            |    | (2)             | (3)                                | (4)         |
| Sequence VIII  | b) | Shear Stability | Refer Note 3                       | ASTM D6709  |

(Page 21, Table 11.1) — Substitute the following for the existing:

**‘Table 11.1 Engine Test Sequence for Petrol Engine Oils  
for EPL6 and EPL 7**

( Clauses 3.1.2 and 3.4 )

| Test Technique         | Rated or Measured Parameter                               | Primary Performance Criteria           |            |                                                      |            |                                                      | Method of Test (Ref to ASTM) |
|------------------------|-----------------------------------------------------------|----------------------------------------|------------|------------------------------------------------------|------------|------------------------------------------------------|------------------------------|
|                        |                                                           | EPL 6                                  |            | EPL 7                                                |            | EPL 7 with resource conserving                       |                              |
| (1)                    | (2)                                                       | (3)                                    |            | (4)                                                  |            | (5)                                                  | (6)                          |
|                        |                                                           | SAE 0W-20, 5W-20, 0W-30, 5W-30, 10W-30 | All others | SAE 0W-16, 5W-16, 0W-20, 5W-20, 0W-30, 5W-30, 10W-30 | All others | SAE 0W-16, 5W-16, 0W-20, 5W-20, 0W-30, 5W-30, 10W-30 |                              |
| <b>Sequence III G</b>  |                                                           | Pass                                   | Pass       | Pass                                                 | Pass       | Pass                                                 | D7320                        |
|                        | Kinematic Viscosity, percent increase at 40°C, <i>Max</i> | 150                                    | 150        | 150                                                  | 150        | 150                                                  |                              |
|                        | Weighted Piston deposit Rating, <i>Min</i>                | 3.5                                    | 3.5        | 4                                                    | 4          | 4                                                    |                              |
|                        | Cam plus lifter wear, µm, <i>Max</i>                      | 60                                     | 60         | 60                                                   | 60         | 60                                                   |                              |
|                        | Hot Stuck Rings                                           | None                                   | None       | None                                                 | None       | None                                                 |                              |
| <b>Sequence III GA</b> | Low temperature viscosity performance                     | Pass                                   | NR         | Refer bench test                                     |            |                                                      | D4684                        |

**Table 11.1 ( Continued )**

| Test Technique      | Rated or Measured Parameter                     | Primary Performance Criteria |        |        |        |                                | Method of Test (Ref to ASTM) |
|---------------------|-------------------------------------------------|------------------------------|--------|--------|--------|--------------------------------|------------------------------|
|                     |                                                 | EPL 6                        |        | EPL 7  |        | EPL 7 with resource conserving |                              |
| (1)                 | (2)                                             | (3)                          |        | (4)    |        | (5)                            | (6)                          |
| <b>Sequence IVA</b> |                                                 | Pass                         | Pass   | Pass   | Pass   | Pass                           | D6891                        |
|                     | Cam wear average, $\mu\text{m}$ , <i>Max</i>    | 90                           | 90     | 90     | 90     | 90                             |                              |
| <b>Sequence VG</b>  |                                                 | Pass                         | Pass   | Pass   | Pass   | Pass                           | D6593                        |
|                     | Average engine sludge rating, <i>Min</i>        | 7.8                          | 7.8    | 8.0    | 8.0    | 8.0                            |                              |
|                     | Rocker arm cover sludge rating, <i>Min</i>      | 8                            | 8      | 8.3    | 8.3    | 8.3                            |                              |
|                     | Average piston skirt varnish rating, <i>Min</i> | 7.5                          | 7.5    | 7.5    | 7.5    | 7.5                            |                              |
|                     | Average engine varnish rating, <i>Min</i>       | 8.9                          | 8.9    | 8.9    | 8.9    | 8.9                            |                              |
|                     | Oil screen clogging, percent, <i>Max</i>        | Report                       | Report | 15     | 15     | 15                             |                              |
|                     | Compression ring sticking (hot stuck)           | NR                           | NR     | None   | None   | None                           |                              |
|                     | Cold stuck rings                                | NR                           | NR     | Report | Report | Report                         |                              |
|                     | Oil screen debris, percent                      | Report                       | Report | Report | Report | Report                         |                              |
|                     | Oil ring clogging, percent                      | Report                       | Report | Report | Report | Report                         |                              |

**Table 11.1 ( Continued )**

| Test Technique | Rated or Measured Parameter                               | Primary Performance Criteria |    |       |    |                                | Method of Test (Ref to ASTM) |
|----------------|-----------------------------------------------------------|------------------------------|----|-------|----|--------------------------------|------------------------------|
|                |                                                           | EPL 6                        |    | EPL 7 |    | EPL 7 with resource conserving |                              |
| (1)            | (2)                                                       | (3)                          |    | (4)   |    | (5)                            | (6)                          |
| Sequence VID   |                                                           | NR                           | NR | NR    | NR | Pass                           | D7589                        |
|                | For SAE XW-16 viscosity grade                             |                              |    |       |    |                                |                              |
|                | a) FEI SUM, <i>Min</i> , percentage                       | NR                           | NR | NR    | NR | 2.8                            |                              |
|                | b) FEI 2, <i>Min</i> , percentage, after 100 hours ageing | NR                           | NR | NR    | NR | 1.3                            |                              |
|                | For SAE XW-20 viscosity grade                             |                              |    |       |    |                                |                              |
|                | FEI SUM, <i>Min</i> , percentage                          | NR                           | NR | NR    | NR | 2.6                            |                              |
|                | FEI 2, <i>Min</i> , percentage, after 100h aging          | NR                           | NR | NR    | NR | 1.2                            |                              |
|                | For SAE X W-30 viscosity grade                            |                              |    |       |    |                                |                              |
|                | FEI SUM, <i>Min</i> , percentage                          | NR                           | NR | NR    | NR | 1.9                            |                              |
|                | FEI 2, <i>Min</i> , percentage, after 100h aging          | NR                           | NR | NR    | NR | 0.9                            |                              |
|                | For SAE 10W-30 and all other viscosity grades             |                              |    |       |    |                                |                              |
|                | FEI SUM, <i>Min</i> , percentage                          | NR                           | NR | NR    | NR | 1.5                            |                              |
|                | FEI 2, <i>Min</i> , percentage, after 100h aging          | NR                           | NR | NR    | NR | 0.6                            |                              |

**Table 11.1 ( Concluded )**

| Test Technique | Rated or Measured Parameter         | Primary Performance Criteria |      |       |      |                                | Method of Test (Ref to ASTM) |
|----------------|-------------------------------------|------------------------------|------|-------|------|--------------------------------|------------------------------|
|                |                                     | EPL 6                        |      | EPL 7 |      | EPL 7 with resource conserving |                              |
| (1)            | (2)                                 | (3)                          |      | (4)   |      | (5)                            | (6)                          |
| Sequence VIII  |                                     | Pass                         | Pass | Pass  | Pass | Pass                           | D6709                        |
|                | Bearing weight loss, mg, <i>Max</i> | 26                           | 26   | 26    | 26   | 26                             |                              |

NOTE — NR implies not required

[Page 22, Table 11.2 Bench Tests of EPL 6, col (1), row 22, Sequence VIII)] — Substitute the following for the existing:

| Characteristics                 | 0W-20, 5W-20, 0W-30, 5W-30, 10W-30                                               | All Other | Method of Test Ref to ASTM |
|---------------------------------|----------------------------------------------------------------------------------|-----------|----------------------------|
| (1)                             | (2)                                                                              | (3)       | (4)                        |
| (Sequence VIII) Shear Stability | Ten hour stripped kinematic viscosity at 100 C, KV must remain in original grade |           | ASTM D6709                 |
| or                              |                                                                                  |           |                            |
| Shear Stability (30 Passes)     | Refer Table 11.3A                                                                |           | ASTM D6278                 |

*for*

| Characteristics                 | 0W-20, 5W-20, 0W-30, 5W-30, 10W-30                                               | All Other | Method of Test Ref to ASTM |
|---------------------------------|----------------------------------------------------------------------------------|-----------|----------------------------|
| (1)                             | (2)                                                                              | (3)       | (4)                        |
| (Sequence VIII) Shear Stability | Ten hour stripped kinematic viscosity at 100 C, KV must remain in original grade |           | ASTM D6709                 |

[Page 23, Table 11.3 Bench Tests of EPL 7, Col (1), Row 26, Sequence viii)] — Substitute the following:

| Characteristics                    | 0W-16, 5W-16, 0W-20, 5W-20,<br>0W-30, 5W-30, 10W-30                                 | All Other | Method of Test<br>Ref to ASTM |
|------------------------------------|-------------------------------------------------------------------------------------|-----------|-------------------------------|
| (1)                                | (2)                                                                                 | (3)       | (4)                           |
| (Sequence VIII) Shear<br>Stability | Ten hour stripped kinematic viscosity at 100 C,<br>KV must remain in original grade |           | ASTM D6709                    |
| or                                 |                                                                                     |           |                               |
| Shear Stability (30 Passes)        | Refer Table 11.3A                                                                   |           | ASTM D 6278                   |

*for*

| Characteristics                    | 0W-20, 5W-20, 0W-30, 5W-30,<br>10W-30                                               | All Other | Method of Test<br>Ref to ASTM |
|------------------------------------|-------------------------------------------------------------------------------------|-----------|-------------------------------|
| (1)                                | (2)                                                                                 | (3)       | (4)                           |
| (Sequence VIII) Shear<br>Stability | Ten hour stripped kinematic viscosity at 100 C,<br>KV must remain in original grade |           | ASTM D6709                    |

[Page 23, Table 11.3 Bench Tests of EPL 7] — Insert the following after the existing Table 11.3:

‘For Shear Stability either of CRC L-38/Sequence VIII / ASTM D6278 (30 passes) shall be used. The following Table 11.3A is applicable for shear stability mentioned in Table 10.1, Table 11.2 and Table 11.3:

**Table 11.3A Requirements of CRC L-38/Sequence VIII / ASTM D 6278  
Tests Stay-in-Grade**

( Clauses 3.1.2, 3.4; Table 10.1, Table 11.2 and Table 11.3 )

| Viscosity<br>Grade | 10 hr. Stripped Kinematic<br>Viscosity@100°C, mm <sup>2</sup> /S, <i>Min</i><br>(L-38/Sequence VIII) | Kinematic Viscosity@100°C, mm <sup>2</sup> /S,<br><i>Min</i> (ASTM D 6278) |
|--------------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| (1)                | (2)                                                                                                  | (3)                                                                        |
| XW-16              | 6.1                                                                                                  | 5.8                                                                        |
| XW-20              | 6.9                                                                                                  | 6.5                                                                        |

**Table 11.3A ( Concluded )**

| <b>Viscosity<br/>Grade</b> | <b>10 hr. Stripped Kinematic<br/>Viscosity@100°C, mm<sup>2</sup>/S, <i>Min</i><br/>(L-38/Sequence VIII)</b> | <b>Kinematic Viscosity@100°C, mm<sup>2</sup>/S,<br/><i>Min</i> (ASTM D 6278)</b> |
|----------------------------|-------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| (1)                        | (2)                                                                                                         | (3)                                                                              |
| XW-30                      | 9.3                                                                                                         | 8.5                                                                              |
| XW-40                      | 12.5                                                                                                        | 11.5                                                                             |
| XW-50                      | 16.3                                                                                                        | 15.0                                                                             |
| XW-60                      | 21.9                                                                                                        | 19.8                                                                             |